

Properties of MLEs

- **Theorem 6.1.2.** Let $X_1, \dots, X_n$ be i.i.d. with pdf $f(x; \theta)$, $\theta \in \Omega$. For a specified function g , let $\eta = g(\theta)$ be a parameter of interest. Suppose $\hat{\theta}$ is the mle of θ . Then $g(\hat{\theta})$ is the mle of $\eta = g(\theta)$.

$\exp(\lambda)$ with pdf $f(x; \lambda) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$.

$\exp(\theta)$ with pdf $f(x; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$, $x > 0$, $\theta > 0$.

- **Theorem 6.1.3.** Assume R0-R2. Then the likelihood equation

$$\frac{\partial L(\theta)}{\partial \theta} = 0$$

or

$$\frac{\partial l(\theta)}{\partial \theta} = 0$$

has a solution $\hat{\theta}_n$ such that $\hat{\theta}_n \xrightarrow{P} \theta_0$.

- Let $X_1, \dots, X_n$ be random sample from a $N(0, \sigma^2)$ distribution,
 - (a) What is the MLE of σ^2 .
 - (b) What is the MLE of σ .
 - (c) Are they consistent estimators?

